SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Perform dry run testing to trace the execution of algorithms, analyze intermediate steps, and verify the correctness of processes. (BL3)*
Assessment Tool Weightage: 40%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

DRY RUN TEST

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 1: Dry Run of A Search Algorithm (10 Marks)*

Consider the following graph.

```
          (A)
        /     \
    2 /         \ 4
     /            \
    (B)---3-----(C)
     |   \           |
   7 |    \2         | 3
     |      \          |
    (D)---2------(E)
       \
         \2
          (G)
```

Heuristic Values

Node	**h(n)**
A	7
B	6
C	4
D	3
E	2
G	0

Task:

Perform a **dry run of the A* Search algorithm** from **A** to **G**.

For each iteration, show:

- Current node
- Open List
- Closed List
- g(n)
- h(n)
- f(n) = g(n) + h(n)

Finally,

1. Identify the optimal path.
2. Calculate the total path cost.

Question 2: Dry Run of Minimax with Alpha-Beta Pruning (10 Marks)

Consider the following game tree.

```
                 MAX
                /     \
           MIN       MIN
          /   |   \   /   |   \
         3    5    6 9    1    2
```

Task:

Perform a **dry run of the Minimax algorithm with Alpha-Beta Pruning** by evaluating the tree **from left to right**.

For each node, show:

- Alpha (α)
- Beta (β)
- Selected value
- Any node that gets pruned

Finally,

1. Determine the best move for MAX.
2. State the final minimax value.
3. List the pruned nodes (if any).